



Characterization and inference of weighted graph topologies from observations of diffused signals

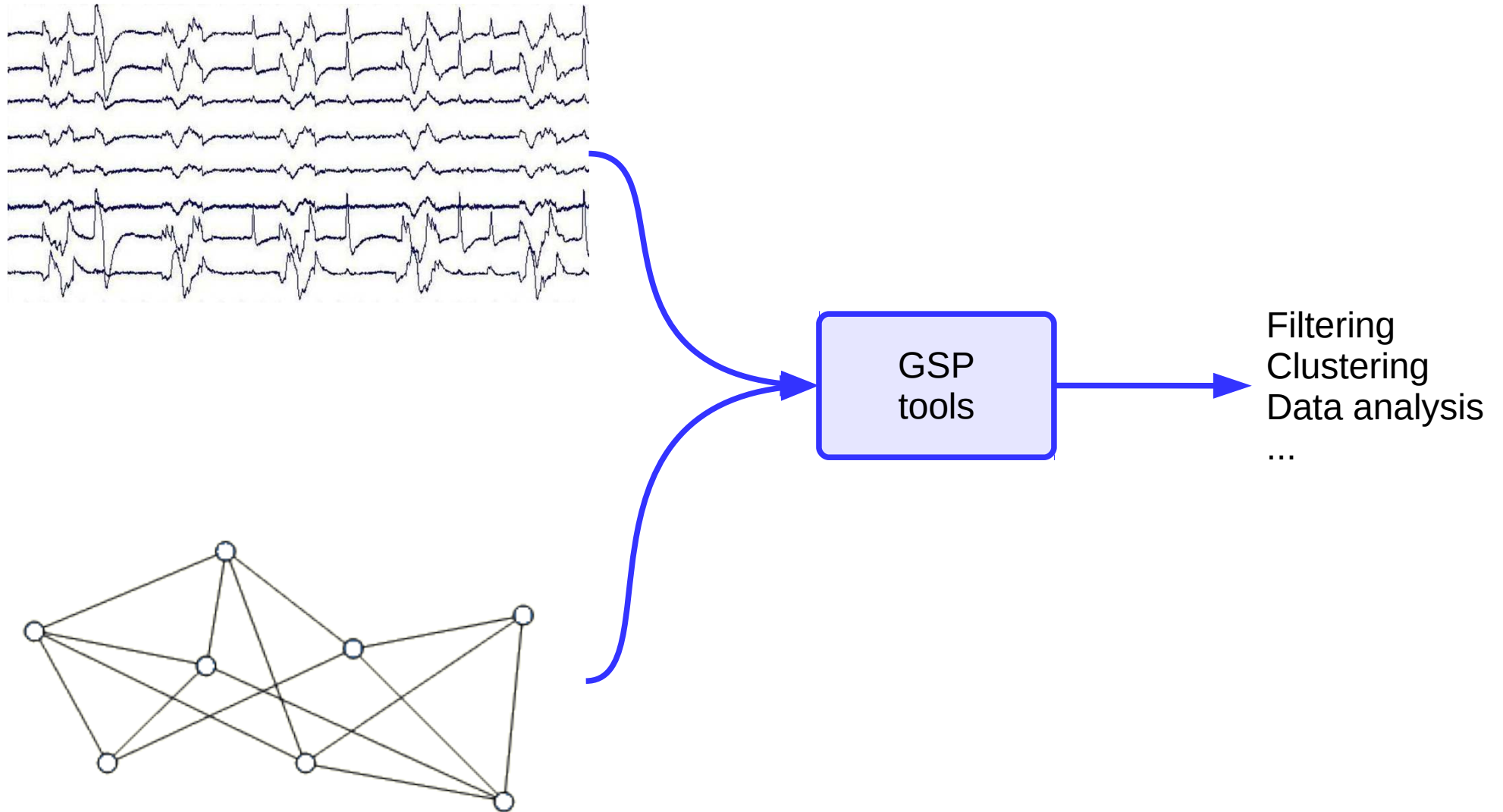
Bastien Padeloup*, Vincent Gripon*, Grégoire Mercier*,
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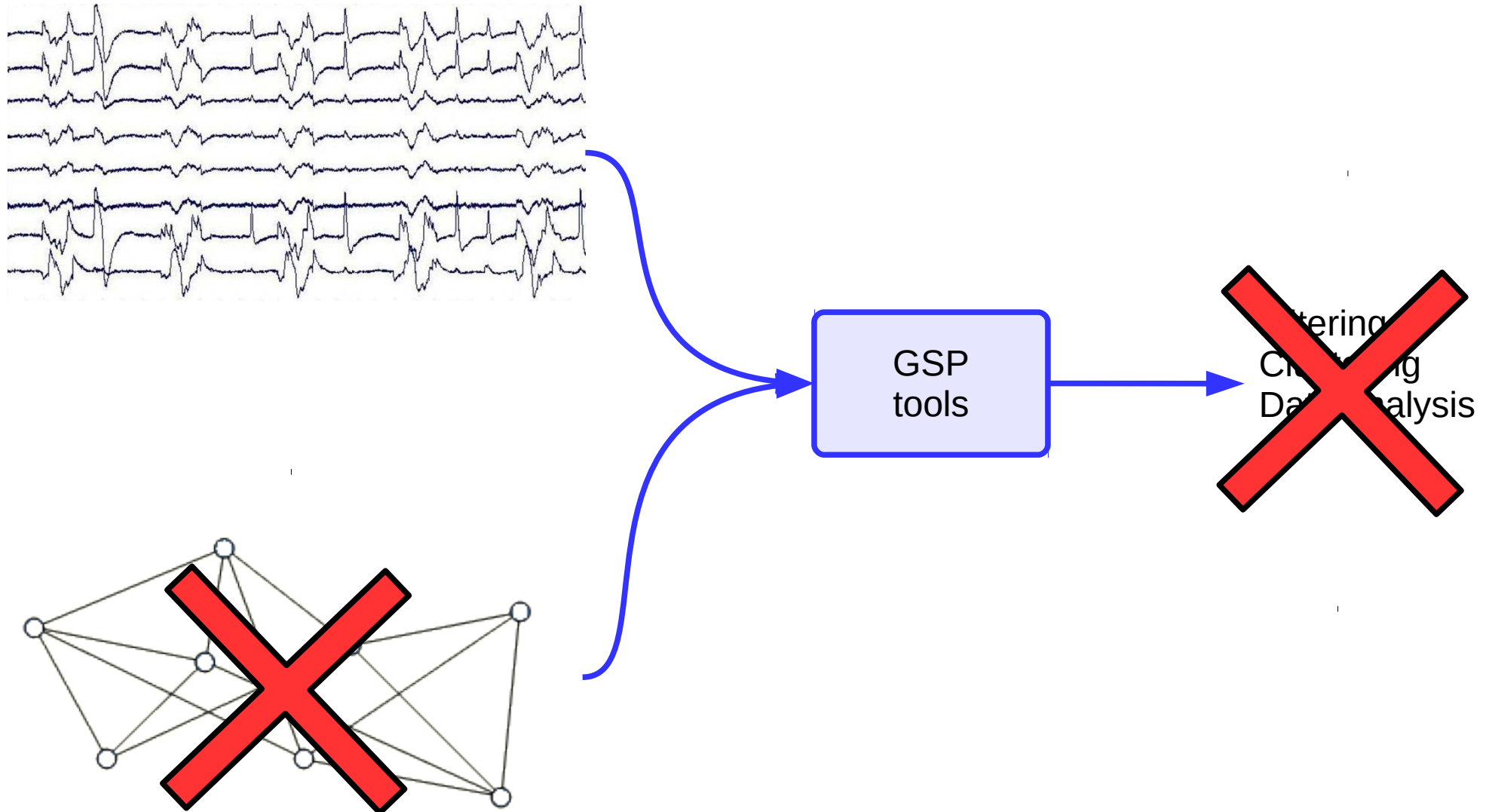
** *name.surname@mcgill.ca*

Preprint version of this work available on arXiv at <http://arxiv.org/abs/1605.02569>

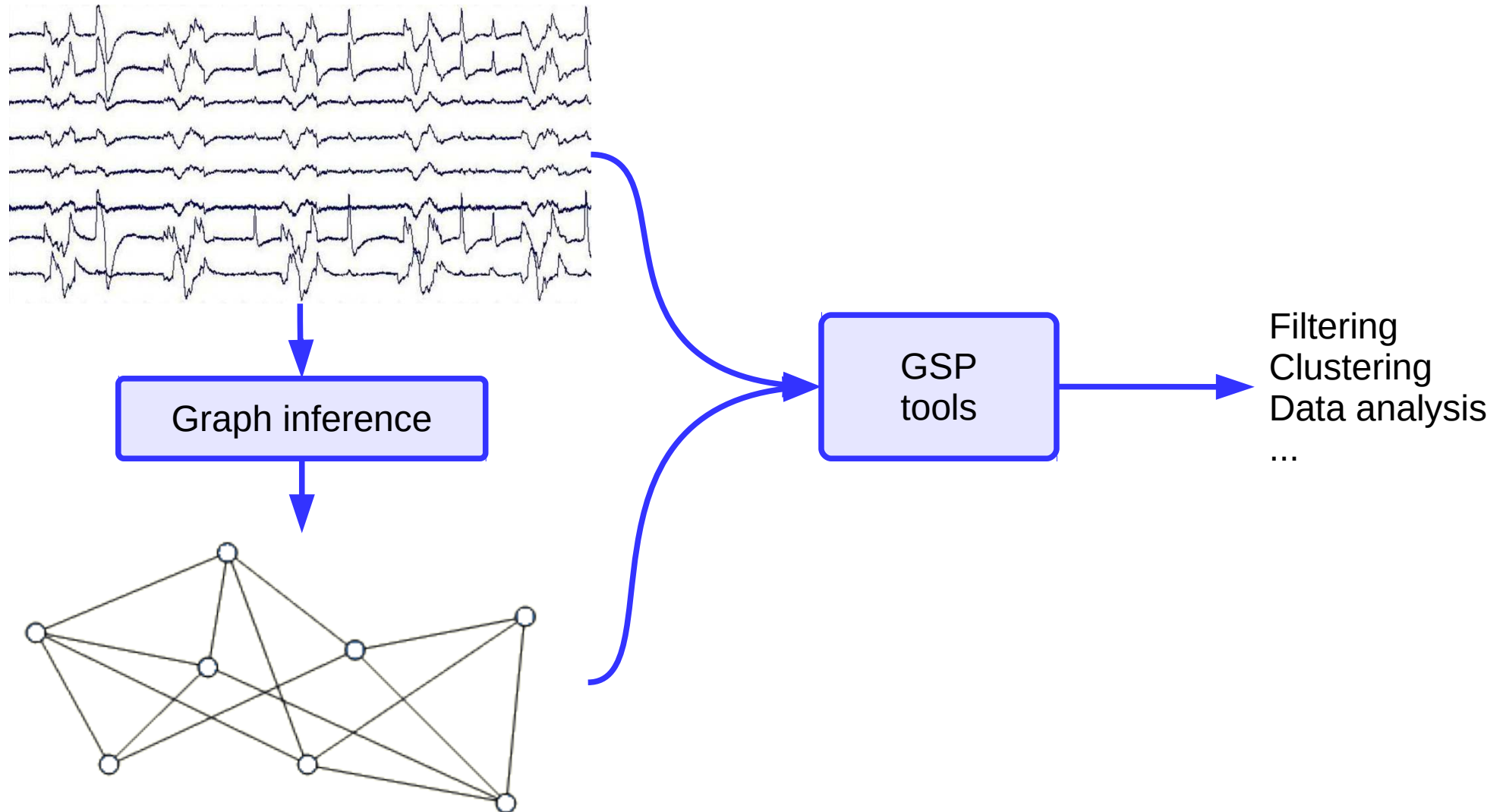
Motivations



Motivations



Motivations



- ▶ Problem formulation
- ▶ A brief overview of related work
- ▶ Characterization of the space of admissible topologies
- ▶ Selecting a point in this space: two example approaches
- ▶ A few results
- ▶ Conclusions & Future work

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General idea of our work

Problem formulation – Related work – Characterization – Selection – Results – Conclusion

- General idea:** Signals observed on a graph have common properties.
These properties should be explainable by the graph topology.
- Translation:** Given a matrix $\mathbf{X}$ of signals, there should exist independent signals $\mathbf{Y}$ such that $\mathbf{X}$ is issued from the evolution of $\mathbf{Y}$ on the graph.

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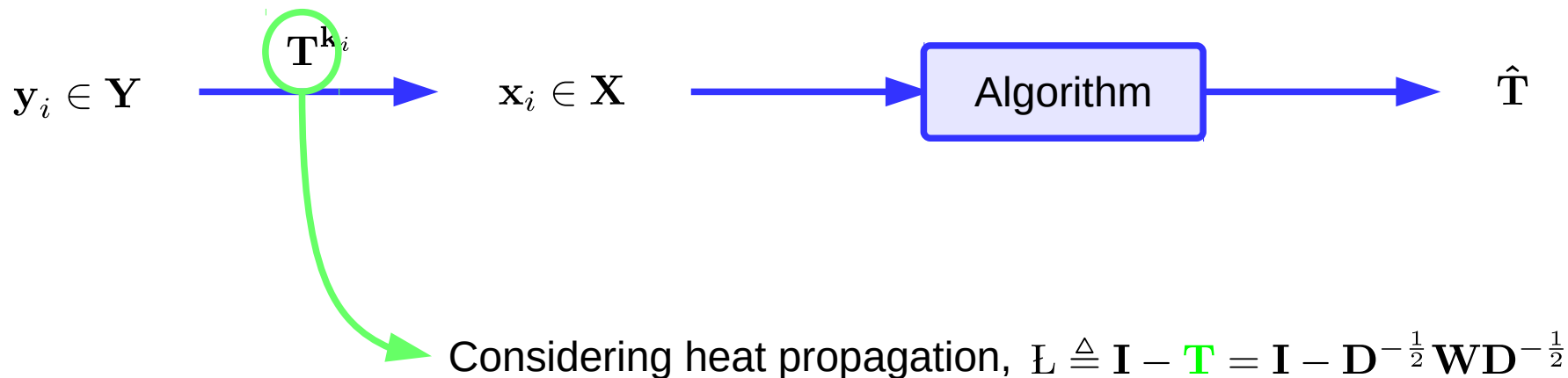
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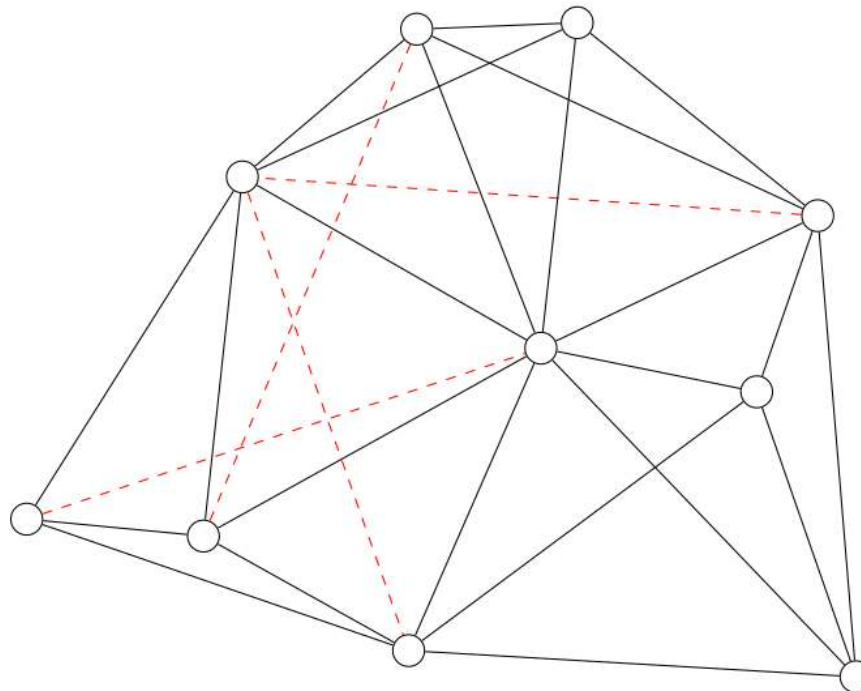
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Covariance thresholding

Problem formulation – **Related work** – Characterization – Selection – Results – Conclusion

General idea: Covariance matrix is a good tool to model relations among nodes. Some of its entries are spurious, let us remove them using a threshold.

Drawbacks: Need for a proper threshold.
Does not eliminate all indirect covariances.



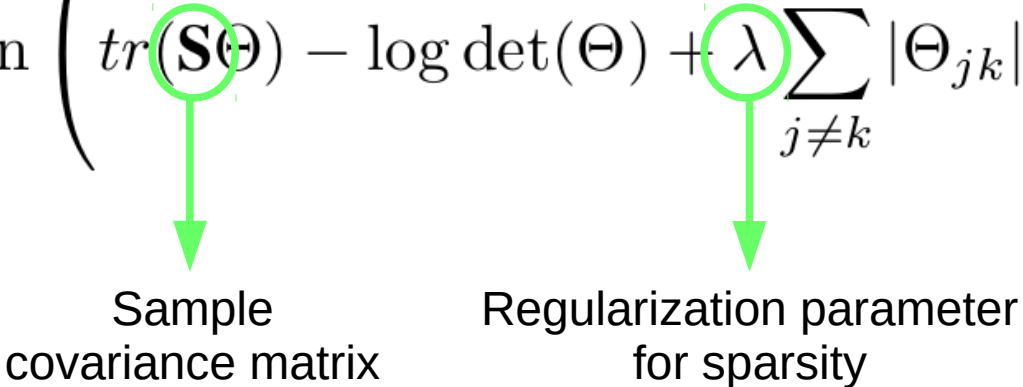
Graphical lasso based methods

Problem formulation – **Related work** – Characterization – Selection – Results – Conclusion

General idea: Sparse inverse of the covariance matrix removes the partial correlations.

Drawbacks: Existence of negative entries in the result matrix.
More a *functional* approach than a *structural* one.

$$\hat{\Theta} = \operatorname{argmin}_{\Theta \geq 0} \left(\operatorname{tr}(\mathbf{S}\Theta) - \log \det(\Theta) + \lambda \sum_{j \neq k} |\Theta_{jk}| \right)$$



Sample covariance matrix Regularization parameter for sparsity

Friedman, Hastie, Tibshirani, *Sparse inverse covariance estimation with the graphical lasso*, Biostatistics, 2008.
Dempster, *Covariance Selection*, Biometrics, 1972.

Smoothness based methods

Problem formulation – **Related work** – Characterization – Selection – Results – Conclusion

General idea: The observed signals should be smooth on the graph.

Drawbacks: Seems legit, although the objective is different from ours.
Work from Dong *et. al* will serve as a baseline for comparisons.

$$S(\mathbf{x}) \triangleq \mathbf{x}^\top \mathbf{L} \mathbf{x} = \sum_{\{u,v\} \in \mathcal{E}} \mathbf{W}(u,v) (\mathbf{x}(u) - \mathbf{x}(v))^2$$

Dong, Thanou, Frossard, Vandergheynst, *Learning Laplacian matrix in smooth graph signal representations*, arXiv, 2014.
Kalofolias, *How to learn a graph from smooth signals*, AISTATS, 2016.

General idea: First version of our work, where $k_i = K$.

Drawbacks: Restrictive case: K is supposed to be constant and known.
Study was only made for an infinite number of signals.

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$$\begin{aligned}\Sigma_X &= \mathbb{E} [\mathbf{X}\mathbf{X}^\top] \\ &= \mathbb{E} [\mathbf{T}^K \mathbf{Y}\mathbf{Y}^\top \mathbf{T}^{K^\top}] \\ &= \mathbf{T}^K \mathbb{E} [\mathbf{Y}\mathbf{Y}^\top] \mathbf{T}^{K^\top} \\ &= \mathbf{T}^{2K}\end{aligned}$$

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$$\Sigma_{\mathbf{X}} = \mathcal{X}_{\Sigma} \Lambda_{\Sigma} \mathcal{X}_{\Sigma}^\top$$

$$\mathbf{T} = \mathcal{X}_{\mathbf{T}} \Lambda_{\mathbf{T}} \mathcal{X}_{\mathbf{T}}^\top$$

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$$\begin{aligned}\Sigma_X &= \chi_\Sigma \Lambda_\Sigma \chi_\Sigma^\top \\ &= \\ \mathbf{T} &= \chi_T \Lambda_T \chi_T^\top\end{aligned}$$

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$$\Sigma_X = \mathcal{X}_\Sigma \Lambda_\Sigma \mathcal{X}_\Sigma^\top$$

$$\Lambda_T = \mathbf{S}^{2K} \sqrt{\Lambda_\Sigma}$$

$$\mathbf{T} = \mathcal{X}_T \Lambda_T \mathcal{X}_T^\top$$

with $\mathbf{S}$ diagonal
matrix of signs
(found by solving an
optimization problem)

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What can we do now?

Problem formulation – Related work – **Characterization** – Selection – Results – Conclusion

Problem: Previous method cannot be applied anymore, now that $\mathbf{k}_i$ are unknown.



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$$\begin{aligned}\widehat{\Sigma}_{\mathbf{X}} &= \sum_{i=1}^M \mathbf{X}(i) \mathbf{X}(i)^\top \\ &= \sum_{k \in \mathbf{k}} \sum_{\substack{i \text{ s.t.} \\ \mathbf{k}(i)=k}} \mathbf{X}(i) \mathbf{X}(i)^\top \\ &= \sum_{k \in \mathbf{k}} \sum_{\substack{i \text{ s.t.} \\ \mathbf{k}(i)=k}} \mathbf{T}^k \mathbf{Y}(i) \mathbf{Y}(i)^\top \mathbf{T}^{k^\top} \\ &= \sum_{k \in \mathbf{k}} \mathbf{T}^k \left(\sum_{\substack{i \text{ s.t.} \\ \mathbf{k}(i)=k}} \mathbf{Y}(i) \mathbf{Y}(i)^\top \right) \mathbf{T}^{k^\top}\end{aligned}$$

$$\begin{aligned}\Sigma_{\mathbf{X}} &= \mathbb{E}_{\mathbf{Y}} \left[\sum_{i=1}^M \mathbf{X}(i) \mathbf{X}(i)^\top \right] \\ &= \sum_{k \in \mathbf{k}} \mathbf{T}^k \mathbb{E}_{\mathbf{Y}} \left[\sum_{\substack{i \text{ s.t.} \\ \mathbf{k}(i)=k}} \mathbf{Y}(i) \mathbf{Y}(i)^\top \right] \mathbf{T}^{k^\top} \\ &= \sum_{k \in \mathbf{k}} |\{i, \mathbf{k}(i) = k\}| \mathbf{T}^{2k},\end{aligned}$$

Important part
of the slide :)

Assumption on the matrix to recover

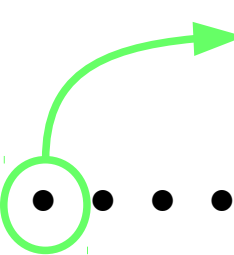
Problem formulation – Related work – **Characterization** – Selection – Results – Conclusion

Current situation: We have the eigenvectors $\mathcal{X}$, let us find the eigenvalues $\hat{\Lambda}$.

Assumption: $\hat{\mathbf{T}}$ should be positive to correspond to a valid graph Laplacian.

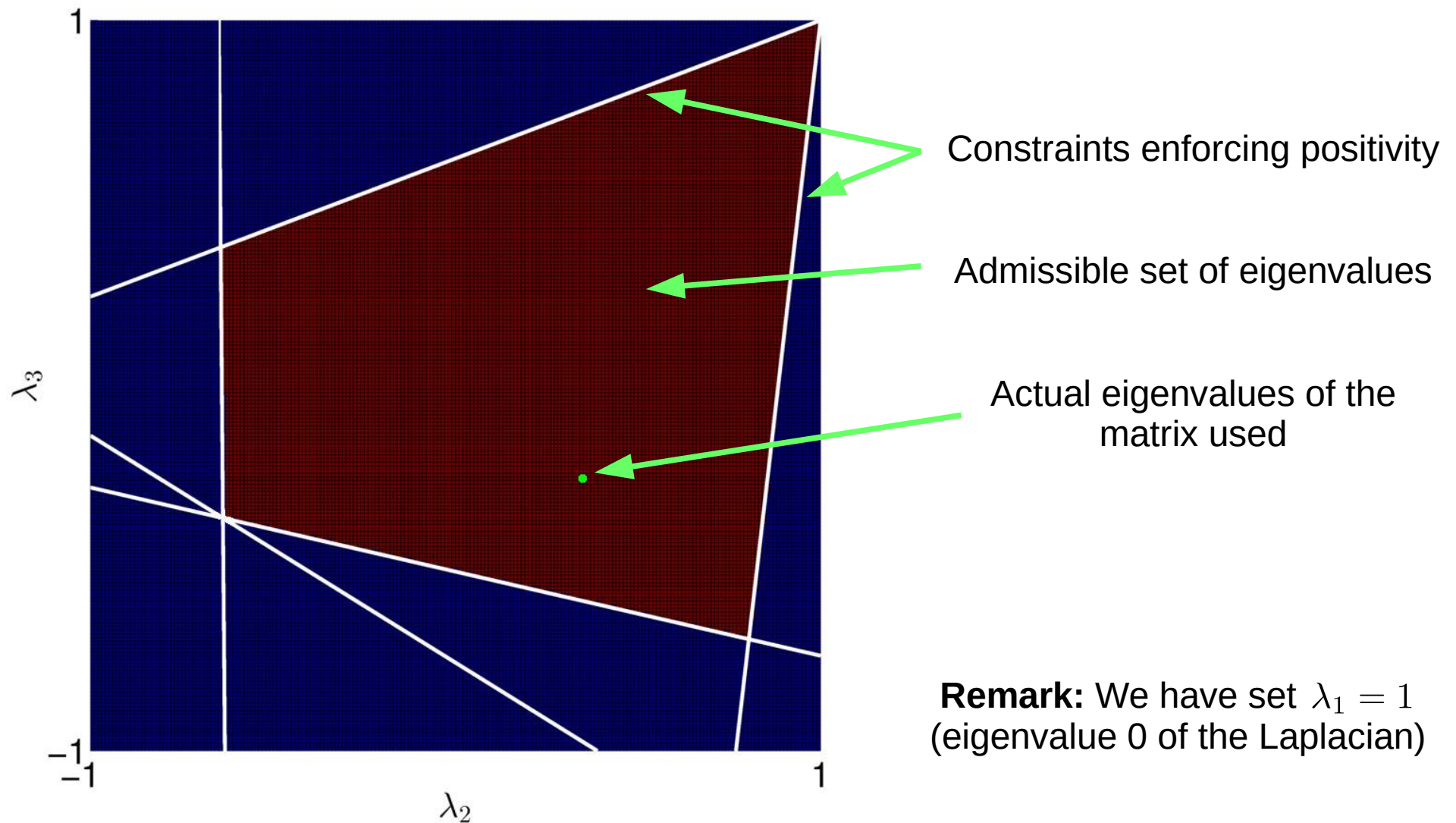
$$\hat{\mathbf{T}} = \begin{pmatrix} \bullet & \bullet & \bullet & \bullet & \bullet & \bullet \\ \bullet & \bullet & \bullet & \bullet & \bullet & \bullet \\ \bullet & \bullet & \bullet & \bullet & \bullet & \bullet \\ \bullet & \bullet & \bullet & \bullet & \bullet & \bullet \\ \bullet & \bullet & \bullet & \bullet & \bullet & \bullet \end{pmatrix}$$

$\hat{\mathbf{T}}(i, j) = \alpha_{ij1} \hat{\lambda}_1 + \alpha_{ij2} \hat{\lambda}_2 + \cdots + \alpha_{ijN} \hat{\lambda}_N$



Example for a 3x3 random matrix

Problem formulation – Related work – **Characterization** – Selection – Results – Conclusion

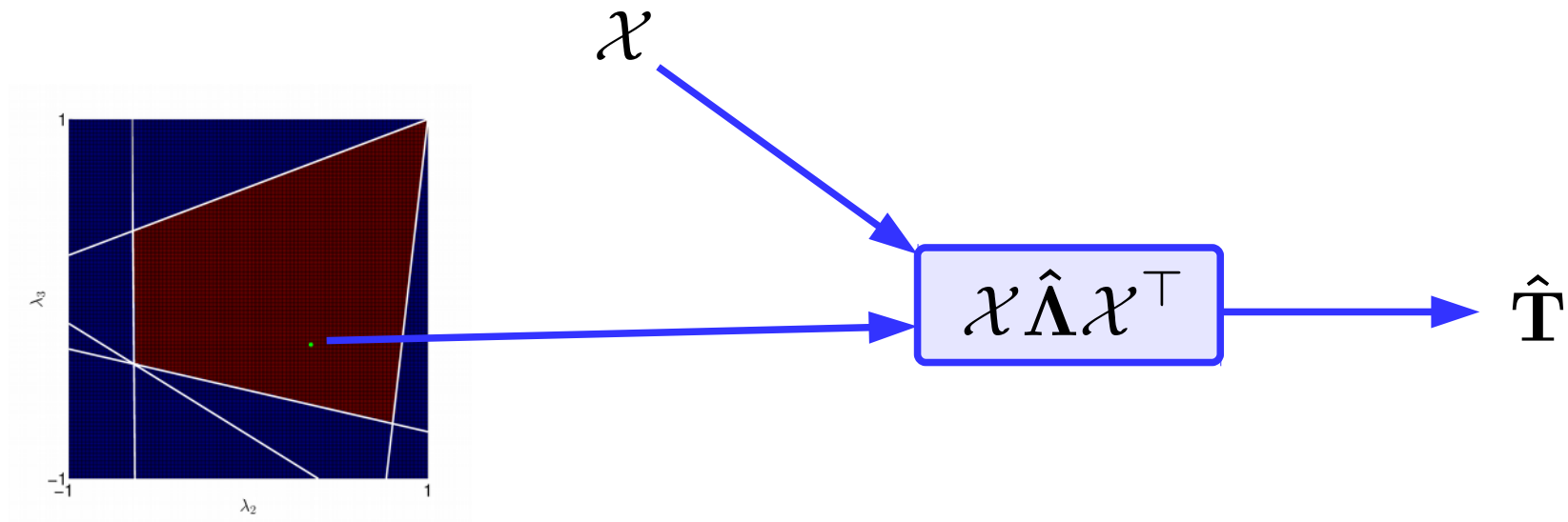


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Which point to choose in this admissible space?

Problem formulation – Related work – Characterization – **Selection** – Results – Conclusion

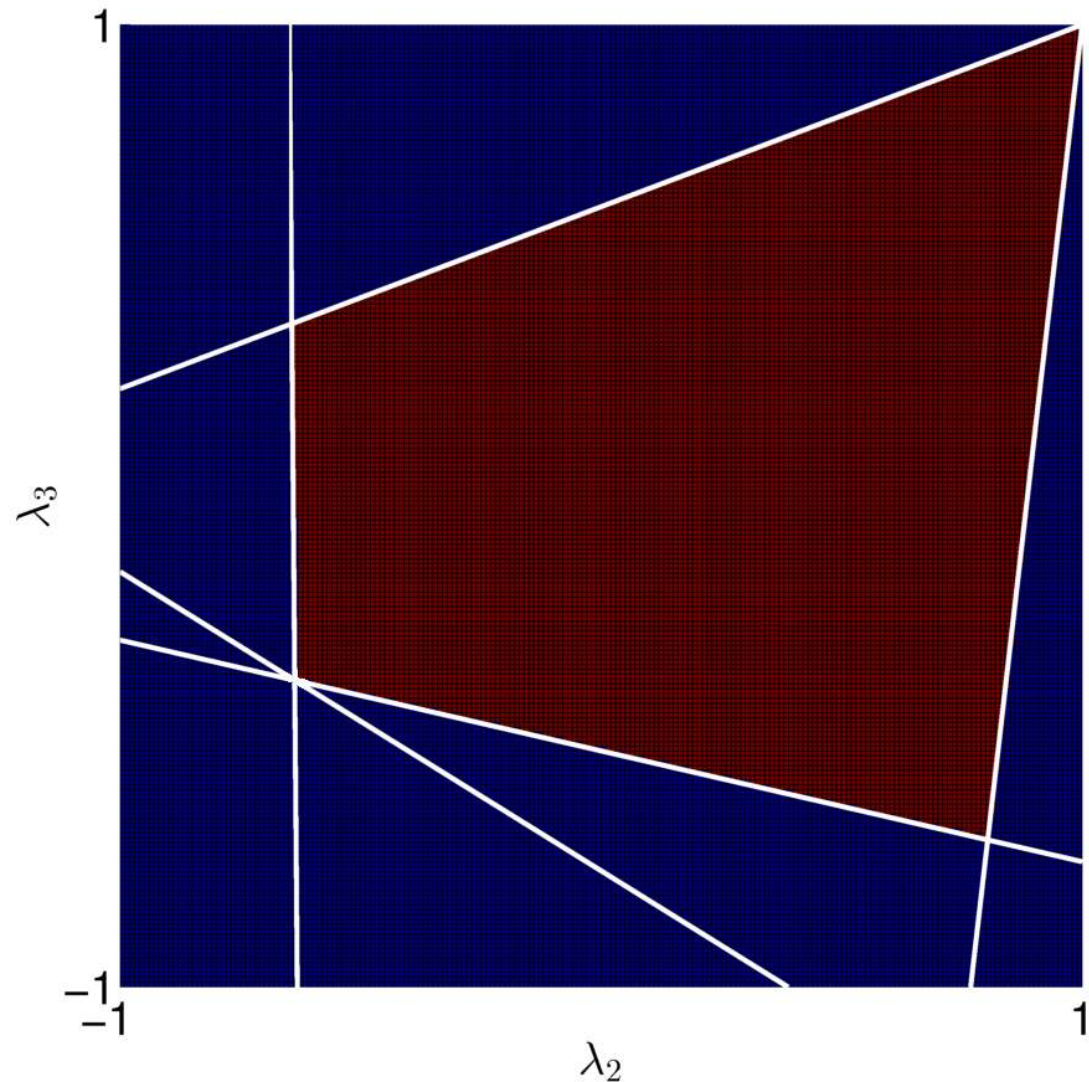
- Enforcing simplicity:** Recovers a matrix with a null diagonal.
Maximises the exchanges between nodes (no conservation).
- Enforcing sparsity:** Common desired property for a graph.
- Others:** Binary entries, distance to a reference graph, smoothness...



Recovering a simple graph

Problem formulation – Related work – Characterization – **Selection** – Results – Conclusion

$$\text{Tr}(\mathbf{T}) = \sum_{i=1}^N \lambda_i$$

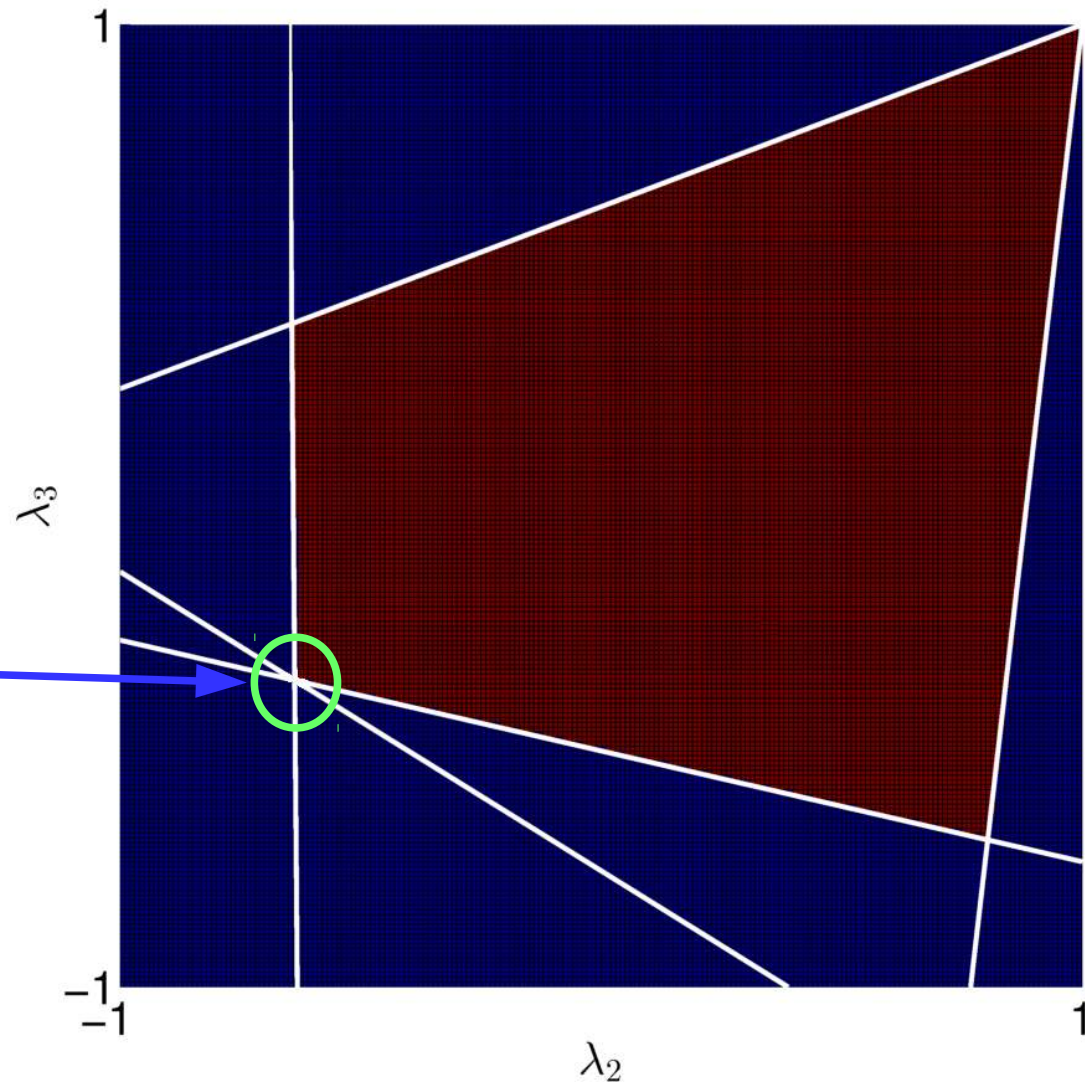


Recovering a simple graph

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Simplex
method

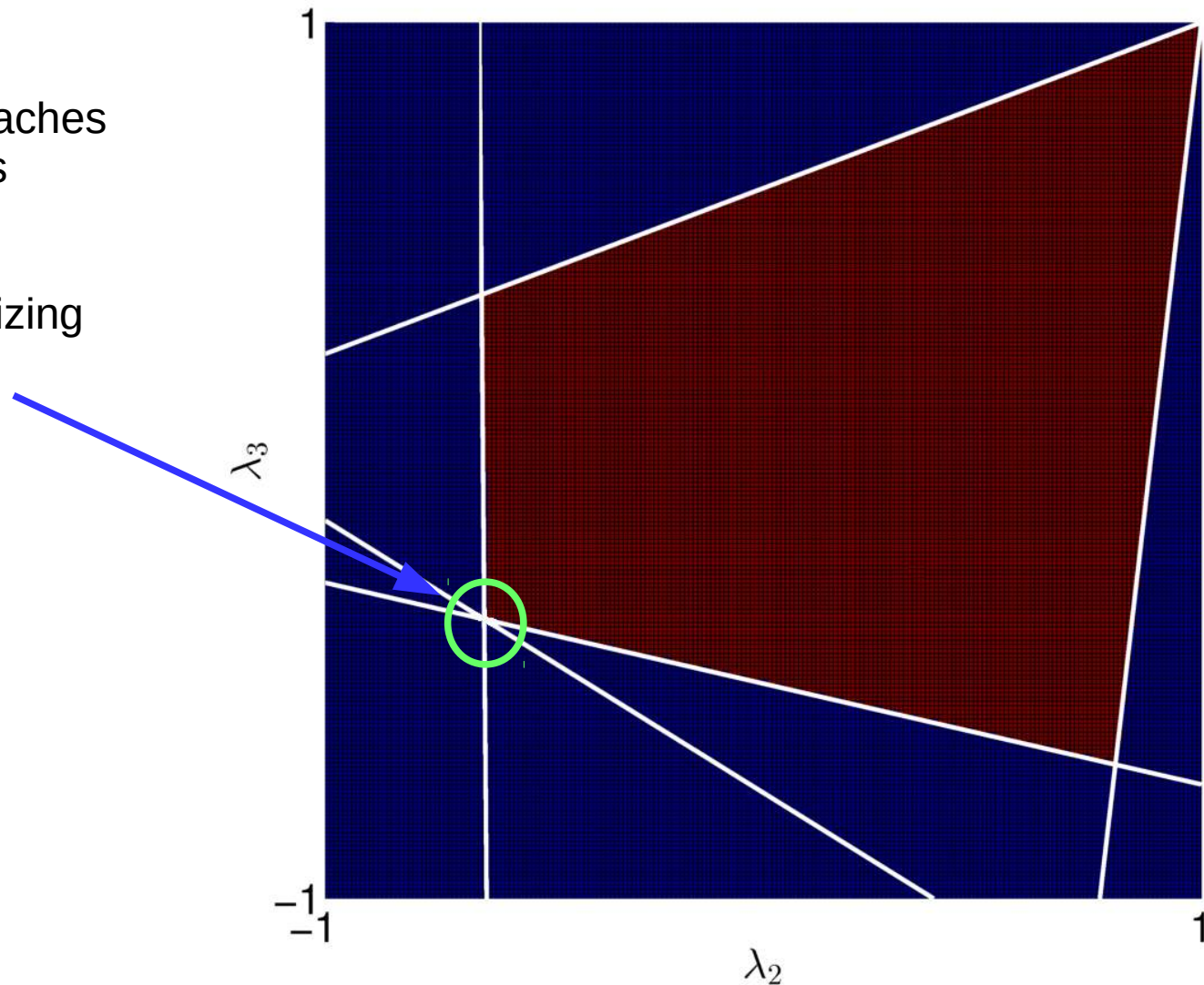


Recovering a sparse graph

Problem formulation – Related work – Characterization – **Selection** – Results – Conclusion

Find the point that reaches
most constraints

Equivalent to minimizing
the l_0 norm



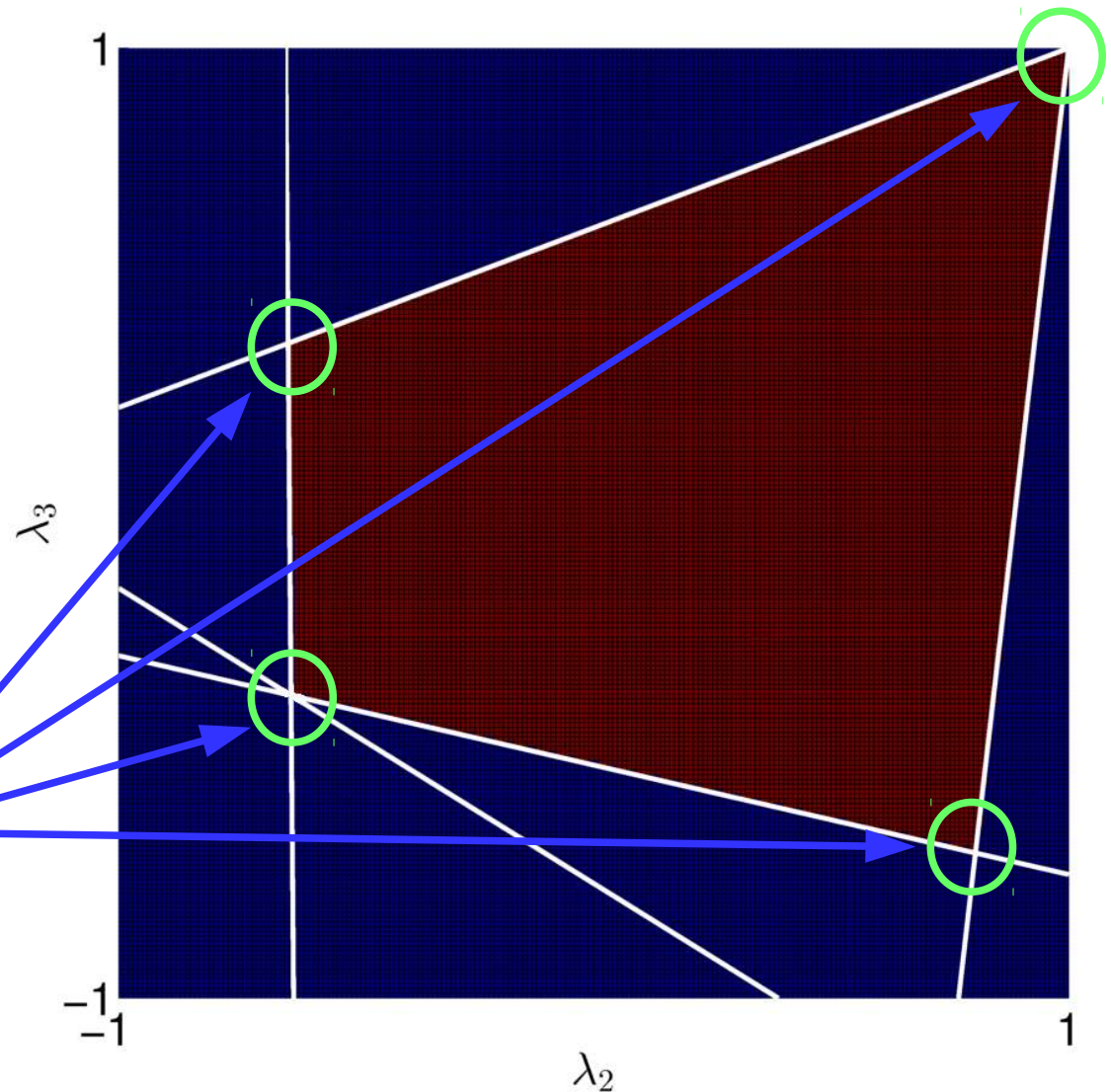
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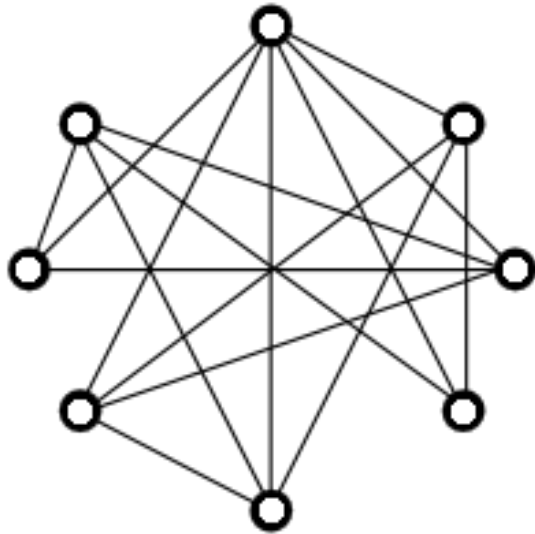
Classical proxy:
Minimize the l_1 norm



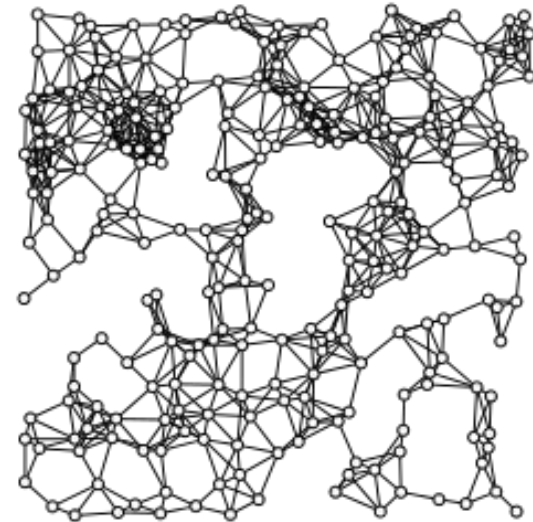
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Experimental assessment of the methods

Problem formulation – Related work – Characterization – Selection – **Results** – Conclusion



Erdős-Rényi random model



Random geometric model



Mean Error Per Reconstructed Entry:

$$\text{MEPRE}(\mathbf{T}, \hat{\mathbf{T}}) \triangleq \frac{2}{N(N-1)} \sum_{i=1}^N \sum_{j=i}^N \left(\frac{\mathbf{T}(i, j)}{\|\mathbf{T}\|_F} - \frac{\hat{\mathbf{T}}(i, j)}{\|\hat{\mathbf{T}}\|_F} \right)^2$$

Reconstruction Error of the Retrieved Eigenvalues:

$$\text{RERE}(\mathbf{T}, \hat{\mathbf{T}}) \triangleq \frac{1}{N} \sum_{i=1}^N \left(\frac{\lambda_i}{\|\mathbf{\Lambda}\|_\infty} - \frac{\hat{\lambda}_i}{\|\hat{\mathbf{\Lambda}}\|_\infty} \right)^2$$

Reconstruction Error of the Powered Retrieved Eigenvalues:

$$\text{REPRE}(\mathbf{T}, \hat{\mathbf{T}}) \triangleq \min_{K \in \mathbb{R}} \frac{1}{N} \sum_{i=1}^N \left(\frac{\lambda_i^K}{\|\mathbf{\Lambda}^K\|_\infty} - \frac{\hat{\lambda}_i}{\|\hat{\mathbf{\Lambda}}\|_\infty} \right)^2$$

Mean Error Per Reconstructed Entry:

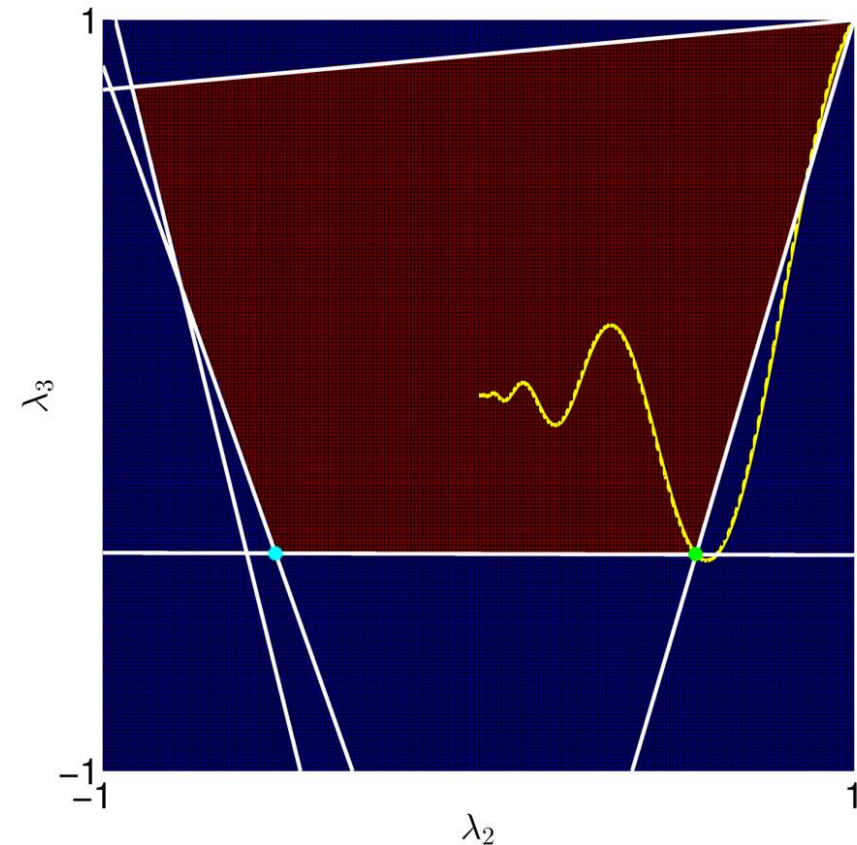
$$\text{MEPRE}(\mathbf{T}, \hat{\mathbf{T}}) \triangleq \frac{2}{N(N-1)} \sum_{i=1}^N \sum_{j=i}^N \left(\frac{\mathbf{T}(i, j)}{\|\mathbf{T}\|_F} - \frac{\hat{\mathbf{T}}(i, j)}{\|\hat{\mathbf{T}}\|_F} \right)^2$$

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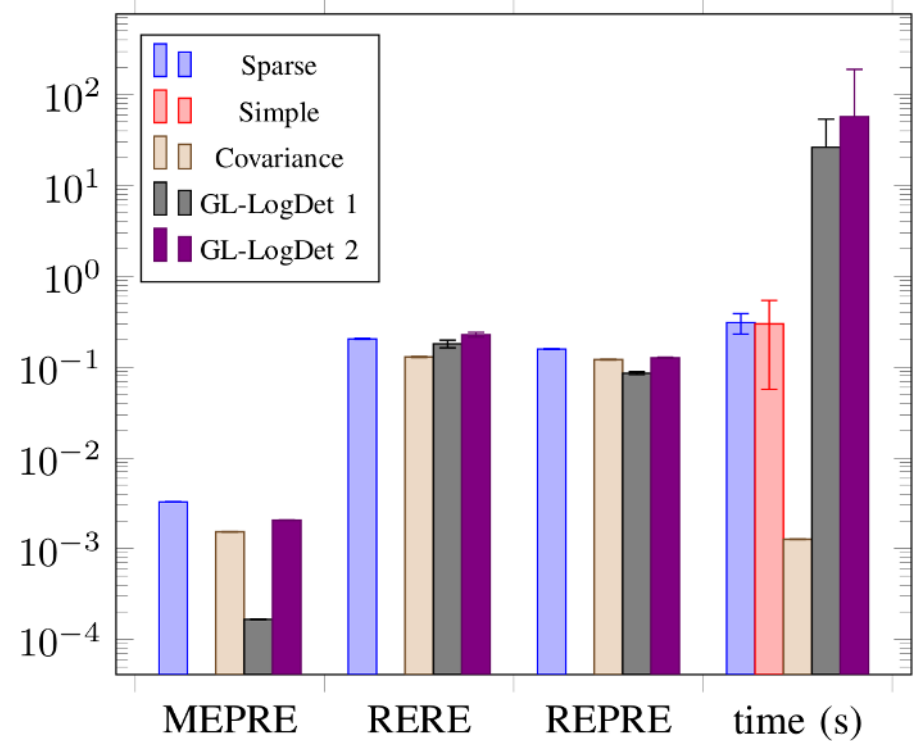
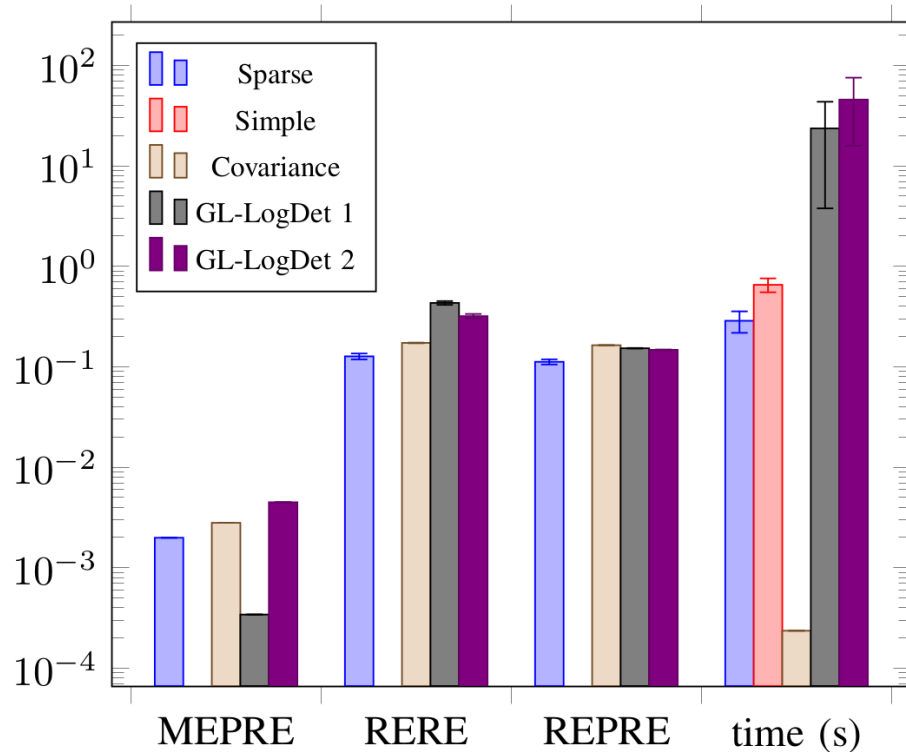
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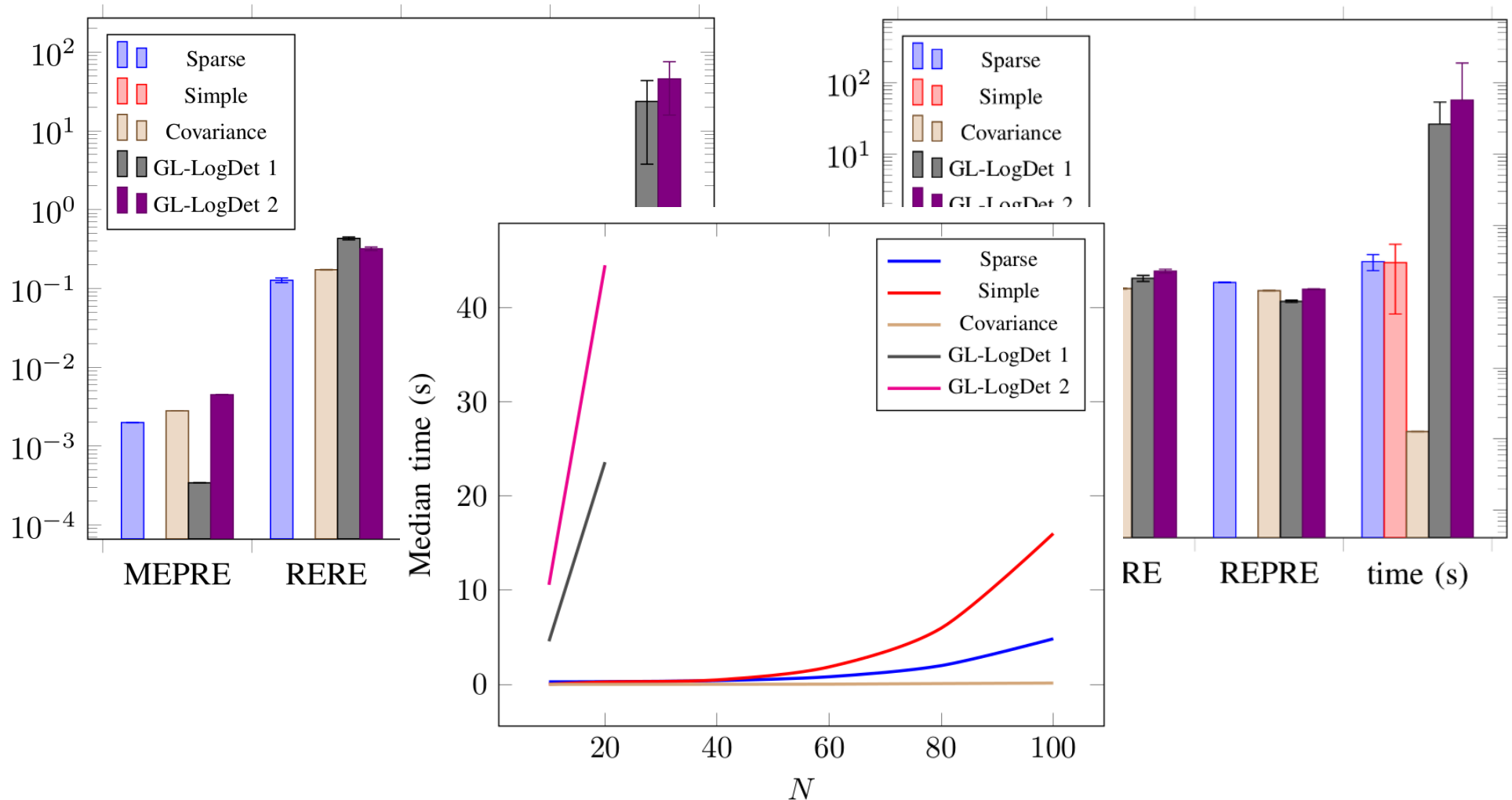
Results when using the groundtruth eigenvectors

Problem formulation – Related work – Characterization – Selection – **Results** – Conclusion



Results when using the groundtruth eigenvectors

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Results when the number of signals is limited

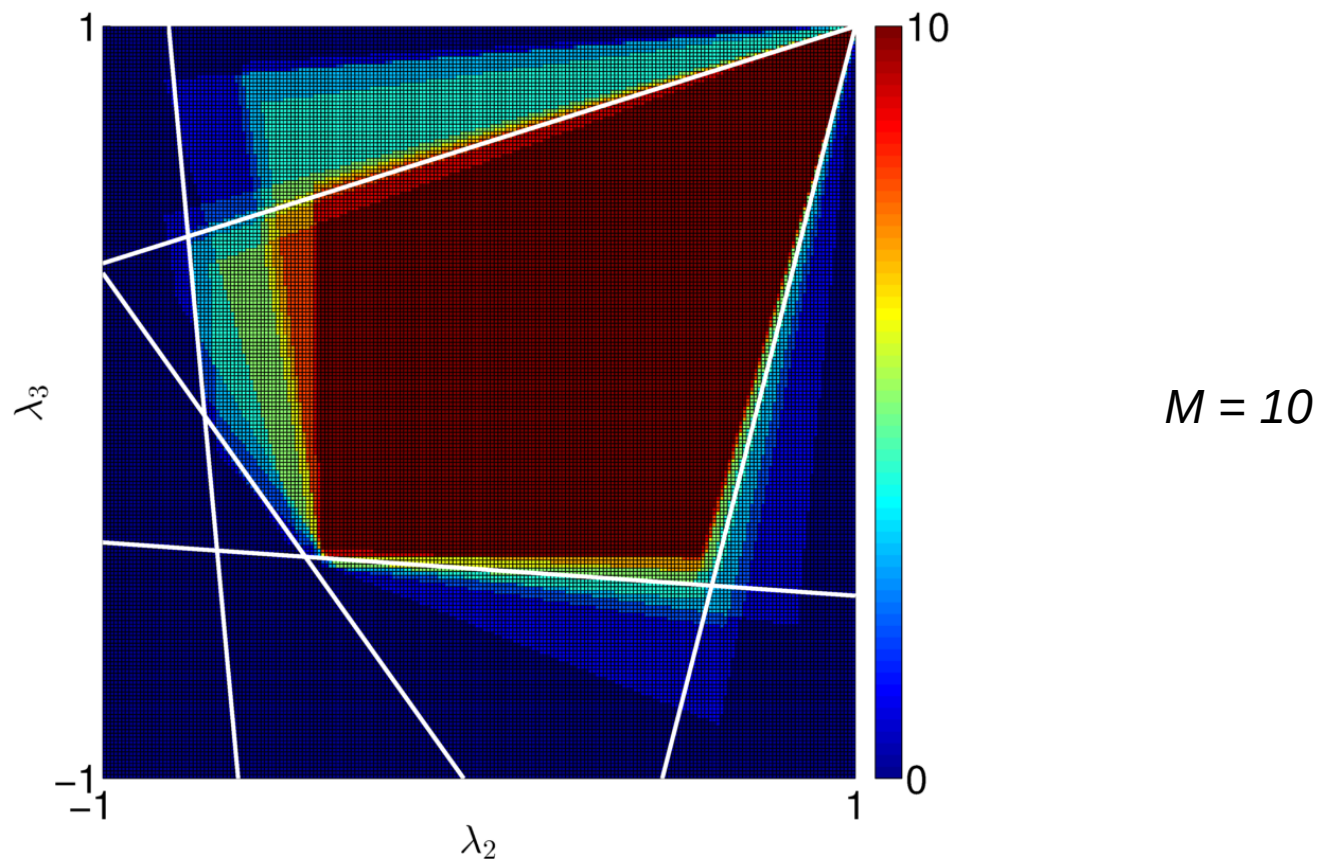
Problem formulation – Related work – Characterization – Selection – **Results** – Conclusion

Impact on the polytope: Perfect retrieval of the eigenvectors is not possible.
PCA provides a good estimator for them.

Results when the number of signals is limited

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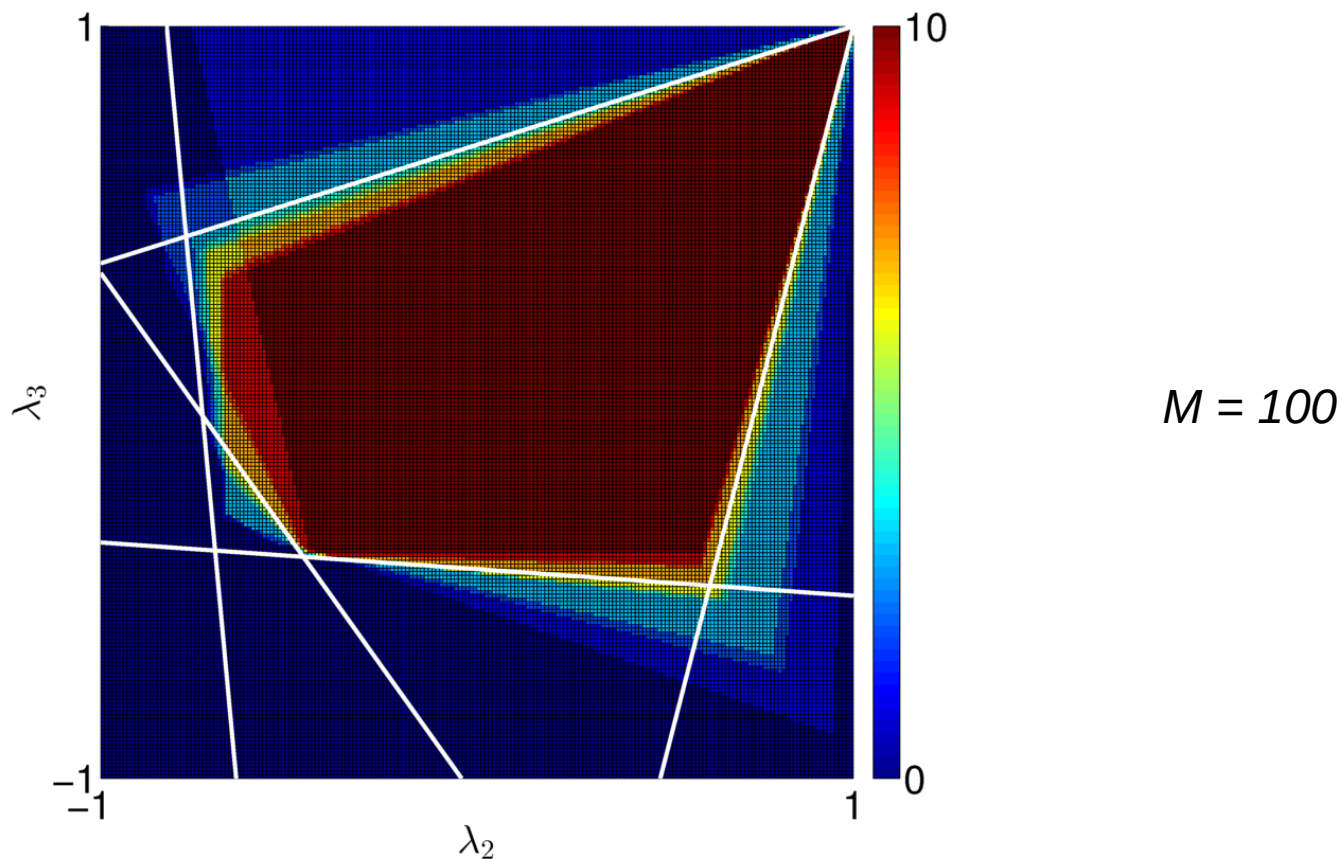


Pearson, *On lines and planes of closest fit to system of points in space*, Philosophical Magazine, 1901.

Results when the number of signals is limited

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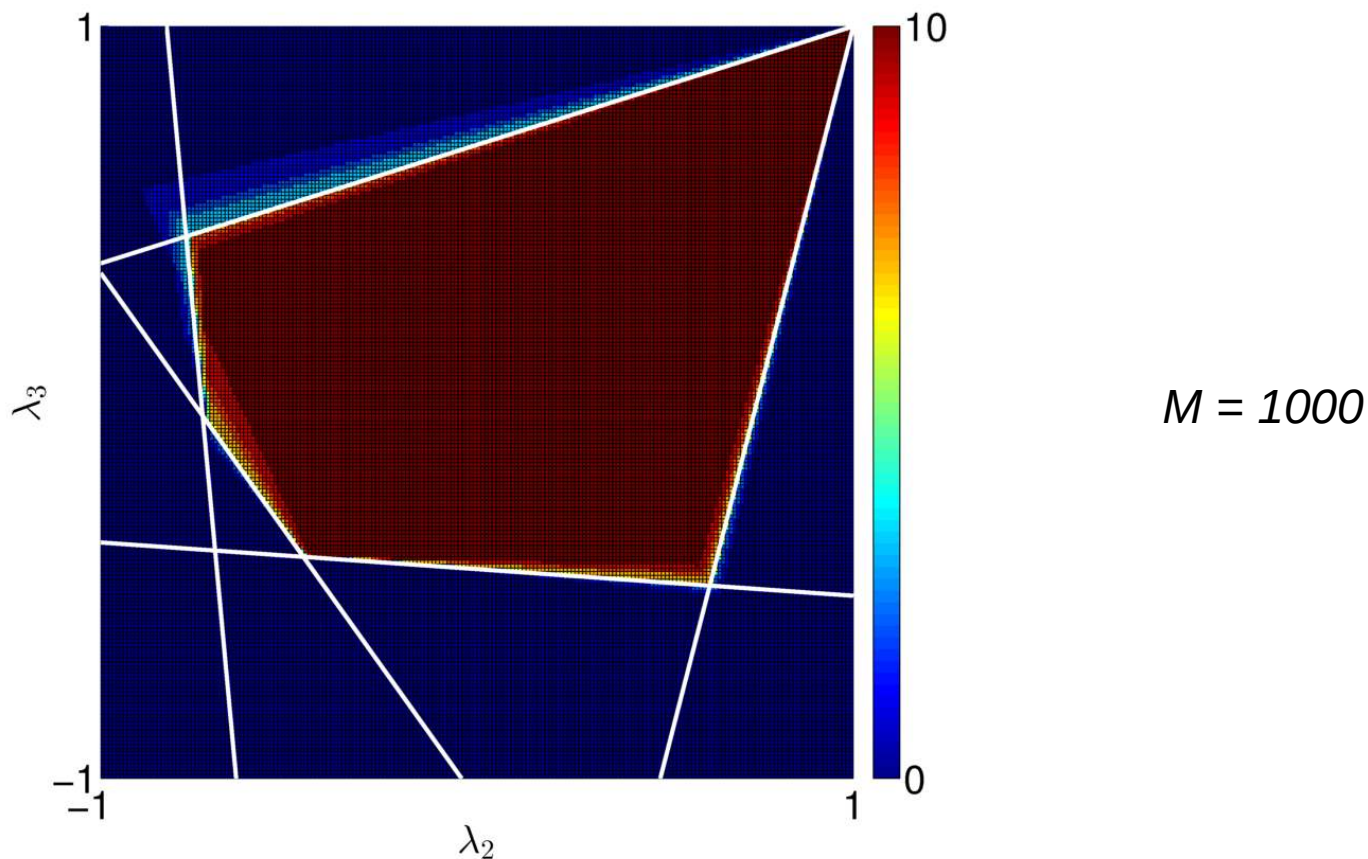


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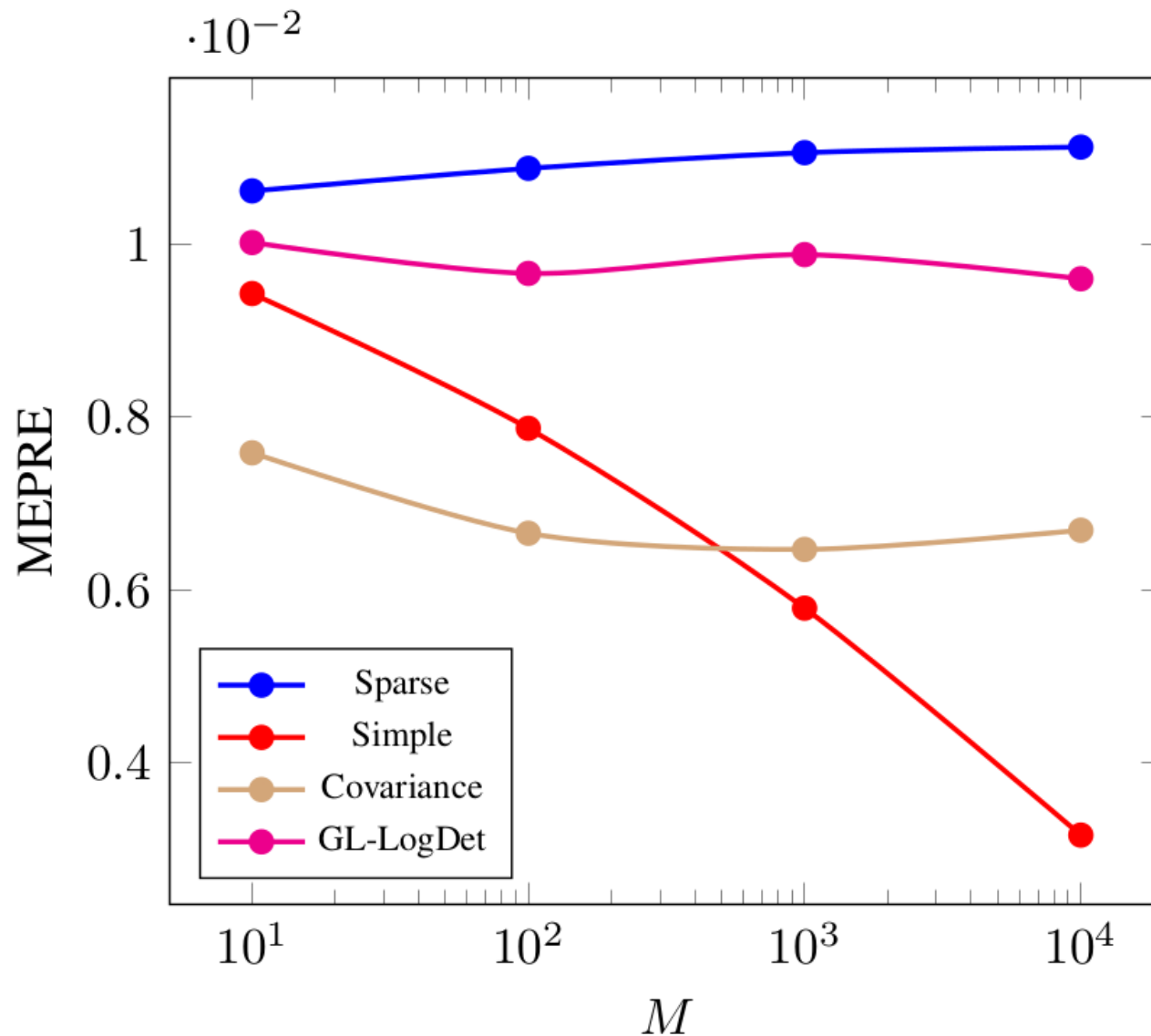
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Convergence towards the actual eigenvectors

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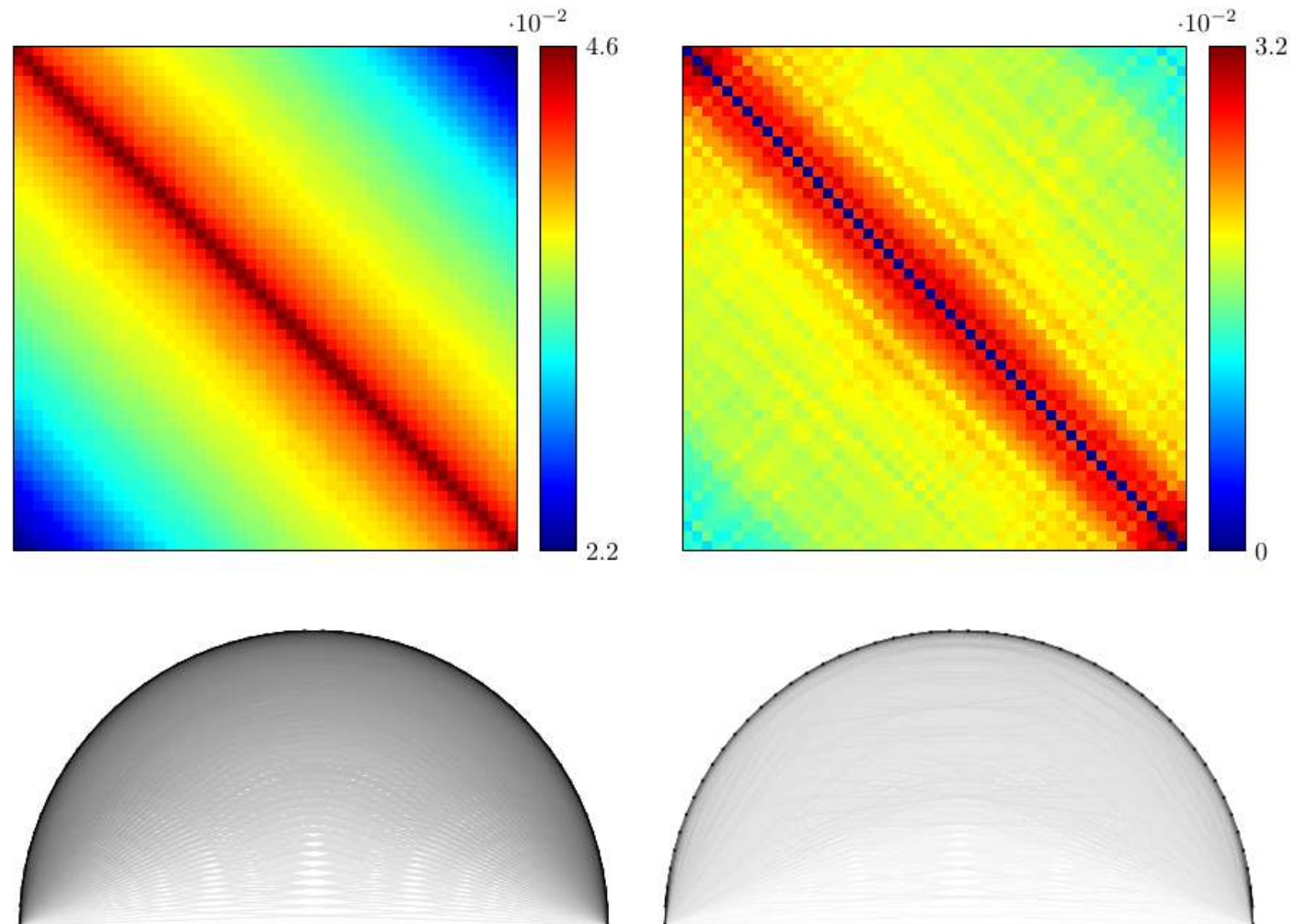


Application to audio signals

Problem formulation – Related work – Characterization – Selection – **Results** – Conclusion



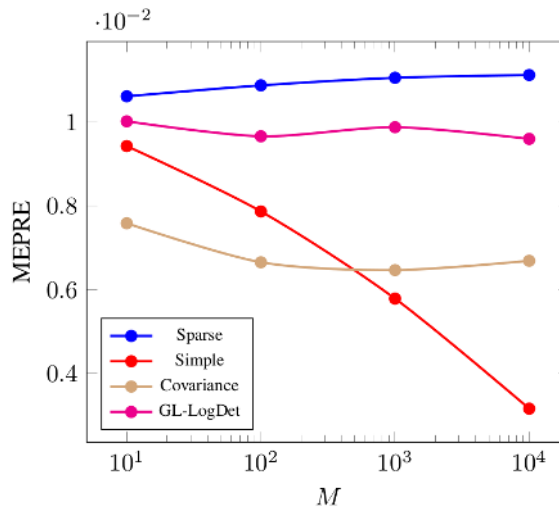
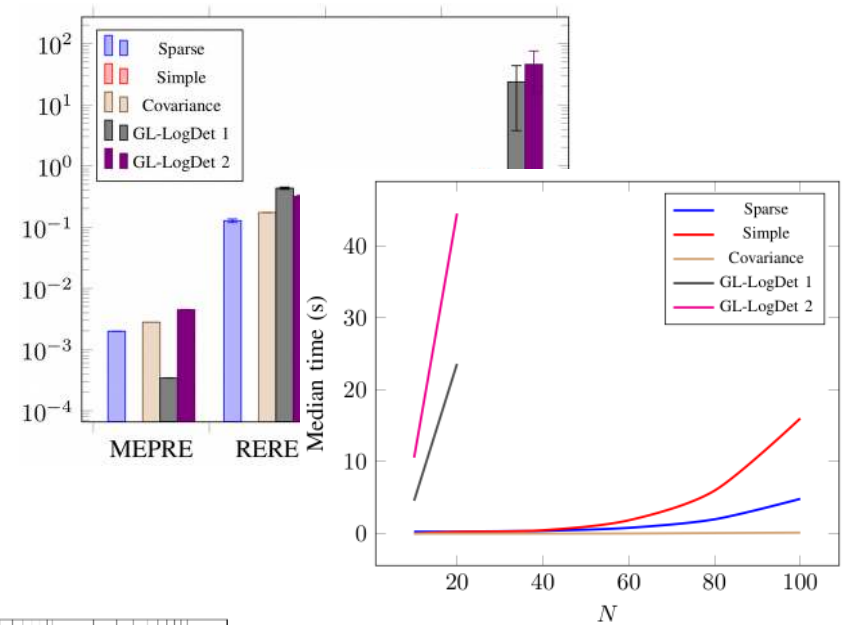
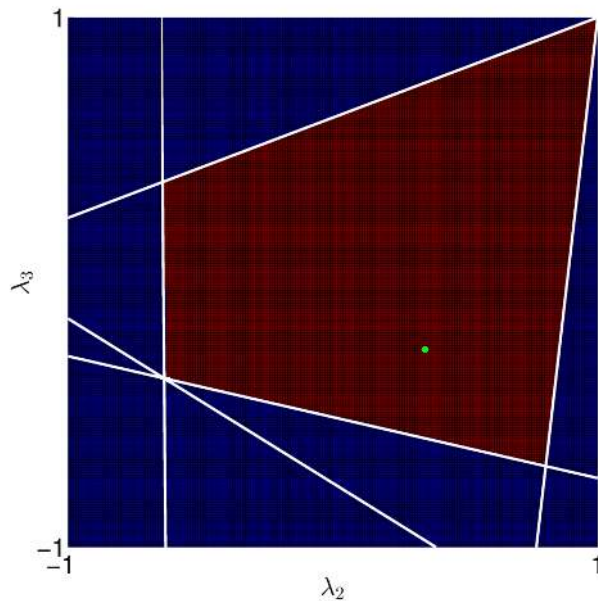
44,100 Hz
 $N = 50$



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Summary of the talk

Problem formulation – Related work – Characterization – Selection – Results – **Conclusion**



- Extensions of this work:**
- Smoothness criterion for the selection of a point?
 - Binarity criterion?
 - Reduce the number of constraints (currently $\frac{N(N-1)}{2}!$)
- Ongoing and close future work:**
- Graph reconstruction for enhanced separability of classes of signals.



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